Introductory Programming Techniques

North Metropolitan TAFE

ICTPRG302: Session 1

Courses:

- Certificate III in Information Technology
- Certificate IV in IT (Programming)
- Certificate IV in IT (Networking)
- Certificate IV in IT (Cybersecurity)
- Certificate IV in IT (Web)

What will I learn?

- Basic programming concepts
- Using Python scripts to solve problems
- Best practices for software development

Background created by Adrian Goold

To complete each week

- Weekly session exercises
- Portfolio questions
- Revision…practise, practise, practise!

Assessments

Assessment Task 1 (AT1): Portfolio

- Weekly Python questions (knowledge and practical exercises)
- Portfolio exercises can be completed in class

Due: Session 4 & 10

Assessment Task 2 (AT2): Project

- Code a simple game
- Developed iteratively in and out of class

Due: Session 18

How to successfully complete this unit

- Complete exercises every week
- Have a playful inquisitive mindset!
  - don't just write code that works, ask why it works?
  - avoid "copy/pasting" from the internet
  - help your classmates: the best way to learn, is to teach
  - don't just give them the answers.

If you want to get good at programming:

practice, practice, practice

and then practice and…

practice some more!

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Additional resources

Freely available online courses, that can be useful to enhance your knowledge.

- Python for Everyone (Py4e)
- LearnPython.org
- W3schools

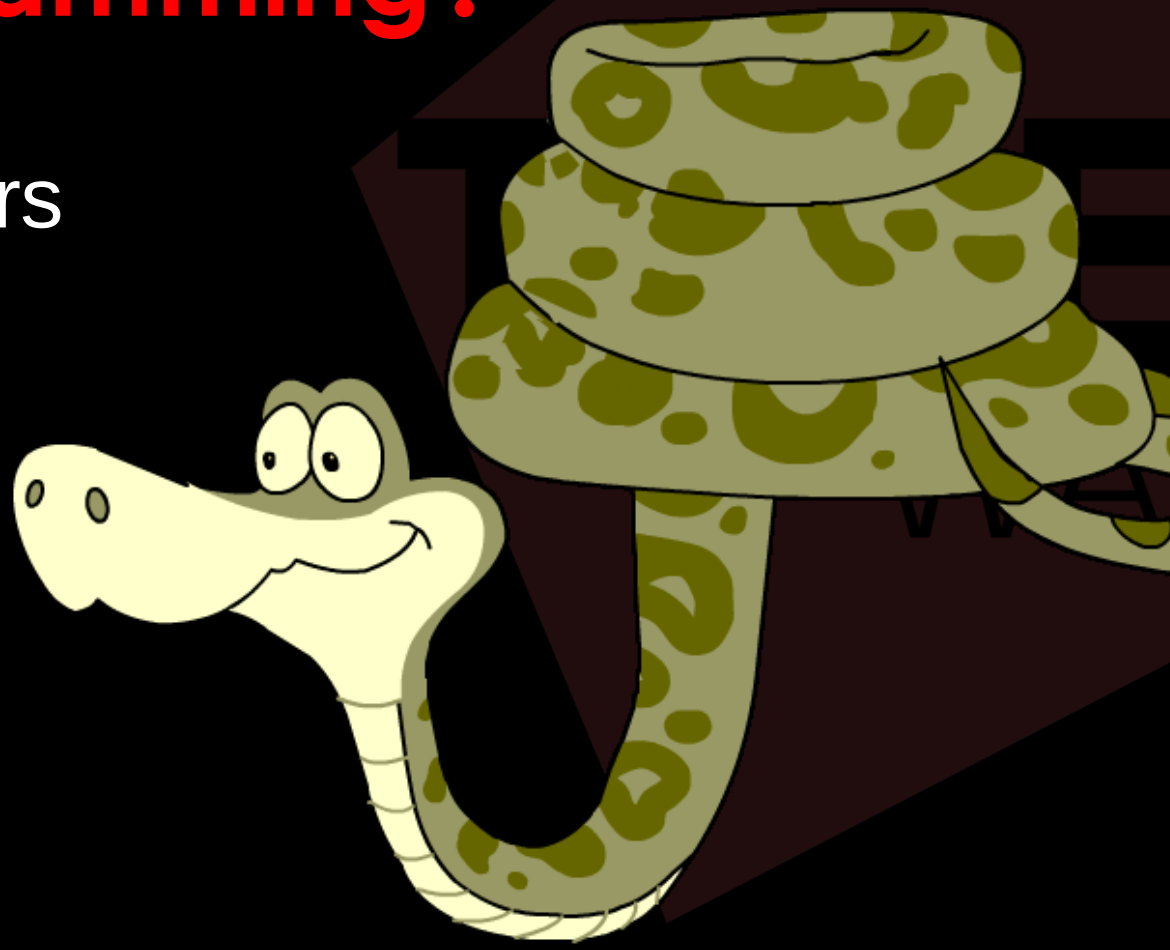
About Blackboard

Blackboard is your online learning platform, containing:

- Weekly session slides
- Resource links
- Assessment information
- Submission links
- Your Grade Centre (assessment results).

What is programming?

- Writing instructions for computers
- Solving complex problems
- Automating tasks
- Building software
- Implementing algorithms
- Enhancing code performance.



Users vs. Programmers

Users

See computers as a set of tools

- word processor
- spreadsheet
- map
- to-do list, etc.

Programmers

Learn the computer “ways”

- use tools to enable them to build new tools
- write code “helpers” for themselves to automate processes.

Why Program?

Can you identify some reasons?

Programming is useful because it…

- Enhances problem-solving skills
- Automates repetitive tasks
- Provide solutions, create applications and games
- Encourages continuous learning
- Skilled programmers are sought after.

Background created by Adrian Goold

What is Code? Software? A Program?

A sequence of stored instructions

- It is a little piece of our intelligence in the computer
- We solve a problem, encode it, and share it to save others time and effort.

A piece of creative art

- Especially when user experience excels.

Background created by Adrian Goold

What is Python?

- Python is a programming language
- Easy to learn
- Used for many applications
- Executes code line by line
- Runs on many platforms
- Supported by a large community.

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Programs for Python...

The clown ran after the car, and the car ran into the tent, and the tent fell down on the clown and the car.

Answer...

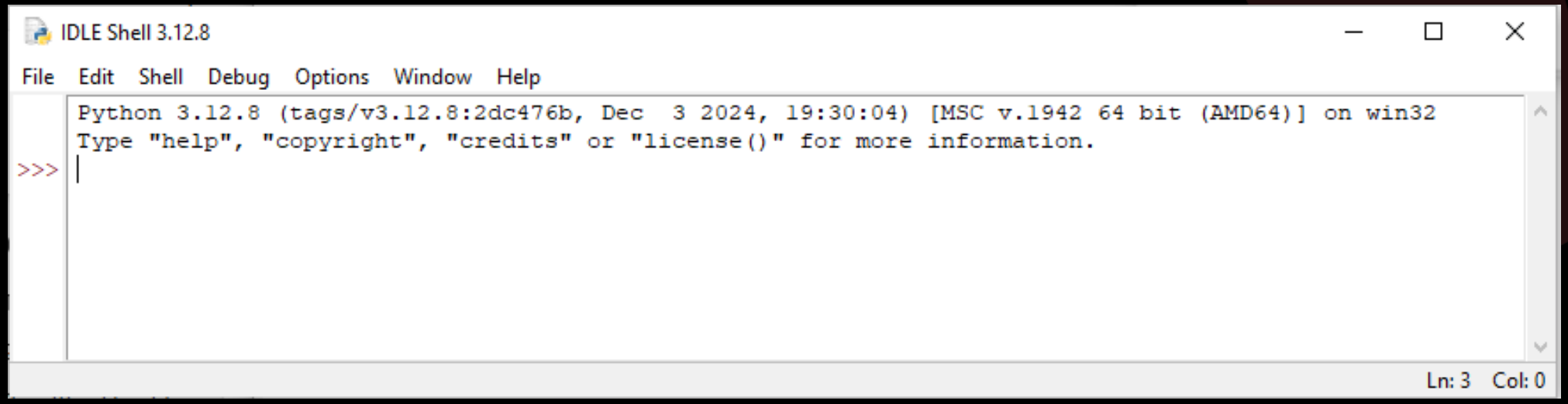
The clown ran after the car, and the car ran into the tent, and the tent fell down on the clown and the car.

Let's code!

What is an IDE?

IDE is an acronym for Integrated Development Environment.

It is a software application for programmers.



What is an IDE?

IDE is an acronym for Integrated Development Environment.

- Offers tools for writing and editing code
- Helps identify and fix errors in code
- Translates code into executable programs
- Organises project files and resources
- Integrates version control systems to track changes in code.

Background created by Adrian Goold

Popular IDE's

IDLE

PyCharm

Visual Studio Code

Background created by Adrian Gould

Python Syntax

- The set of rules that defines how a Python program is written and interpreted.
- Python syntax can be executed by writing directly in the Command Line of an IDE.

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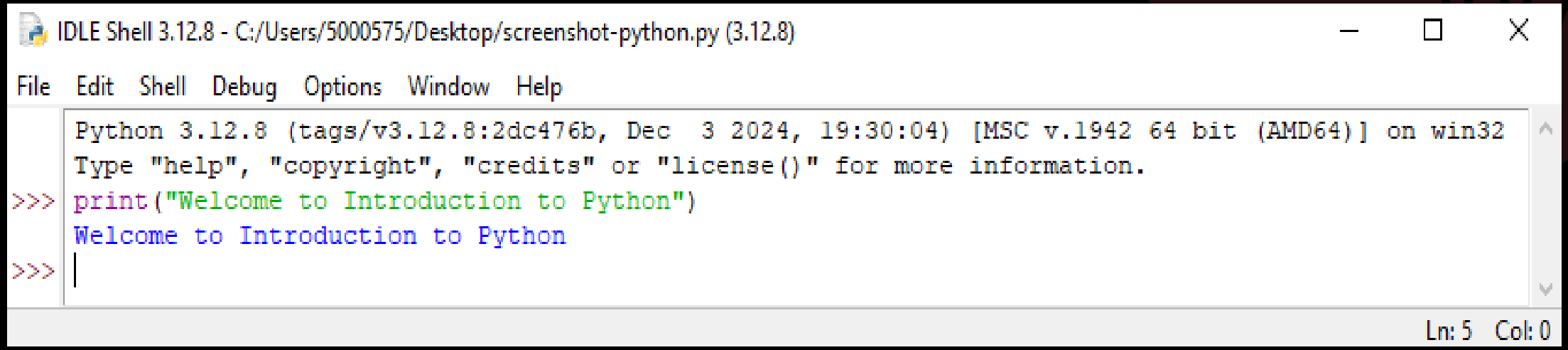
The Command Line

- A text-based interface used to interact with the computer's operating system (OS) by typing commands.
- Unlike graphical user interfaces (GUIs) that use windows, icons and menus, the command line relies on text input and output.
- Users type specific commands to perform tasks such as running programs.

Print Statements

Open **IDLE** and type the following:

print(“Welcome to Introduction to Python”)



More Print Statements

- print(‘Welcome to Introduction to Python’)
- print(“Welcome” + “Mary”)
- print(“Welcome”, “ John”)
- print(Welcome Susan)
- print(“1” + “2”)
- print(1+2)

What have you learned regarding the ***print*** statement?

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Print functions

The Print function displays information on a screen.

print(‘Hello World’)	Displays the text Hello World.
print( )	Text within the brackets will be displayed on screen
“ “ or ‘ ‘	*Text within brackets must be in quotation marks
+	A plus concatenates (joins) *text, no extra space
,	A comma adds a space between *strings

*Note: Text within quotation marks “ “ or ‘ ‘ is referred to as *strings.

Comments in Python

Anything typed after a hashtag # is ignored by Python and will not show on the output screen.

File Edit Format Run Options Window Help

```
# AT3 Project: Guess my Word Game
# Sally Smith 2000000


# select target word at random from TARGET_WORDS
    with open(TARGET_WORDS, "r") as text_file:
        target_word = random.choice(text_file.readlines()).strip()
```

Why comment?

- Describes what is going to happen in a sequence of code
- Documents code authorship and/or other important information
- Turns off a line of code - perhaps temporarily.

Python Coding Standard (PEP8)

PEP 8 is an acronym for Python Enhancement Proposal 8.

- It is a style guide for writing readable Python code
- Ensures uniform coding practices across Python projects
- Aims to make code “clean” and maintainable.

Common PEP 8 Standards:

Code layout:

- Use 4 spaces per indentation level and limit lines to 79 characters.

Naming conventions:

- Use snake_case for variables and functions.

Note: Variables and functions will be explained in further detail in Session 2.

Common PEP 8 Standards:

Python variable names:

- Use lowercase letters: abcdefghijklmnopqrstuvwxyz

Correct:	Incorrect:
file_name	File_Name

Common PEP 8 Standards:

Whitespace:

- Avoid extra spaces around operators (+, -, *,) and after commas.

Comments:

- Write clear, concise comments to describe your code.

Activity to complete

Practise what you have learnt today!

Locate your Week 1 Revision exercises folder and complete the following activities:

1. Mary Had a Little Lamb
2. Song lyrics

Further Practice: Encryption program

1. Open IDLE
2. File / new file
3. Insert a comment:

```
#############################################################
#
# Caesar cipher program
#
# File:      ICTPRG302/CLASSWORK/caesar-cipher.py
# Author:    Ingrid Aldum (ialdum.aldum@nmtafe.wa.edu.au
# Copyright: Copyright 2023, Ingrid Aldum
#############################################################

print("Welcome to my Caesar cipher program")
```

4. Print a welcome to your encryption program

   E.g., Welcome to my Caesar cipher program

5. Save your file and run your program. (Press F5).